How to Kernelize SVM

$$\mathcal{L}_D(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{j=1}^{N} \sum_{i=1}^{N} \alpha_i \alpha_j y_j y_i x_j^T x_i$$

$$w = \sum_{i=1}^{N} \alpha_i y_i x_i$$

$$b = -\frac{1}{2}\left[\max_{i:y_i=-1} w^T x_i + \min_{i:y_i=+1} w^T x_i \right]$$

Class of vector **Z** ⇒ *sign*(W^T Z + b)

Remember :

Mapping Instead of using $\mathbf{X_i}$ (of dimension ***d***) use $\varphi(X_i)$ *(with higher dimension)*

$$X_i^T X_j \Rightarrow \phi(X_i)^T \phi(X_j)$$

Kernelization Instead of using $\varphi(X_i)$, $\varphi(X_j)$ use $\mathbf{K(X_i^T, X_j^T)}$ *(Where K is one of kernel functions)*

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